REFINED PROJECTION: ISO NEAR A SMOOTH POINT OF A SUBVARIETY Z INSIDE X

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B1Q: sources/stacks-chow/tags.txt, source lines 13903–13903. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B1P: sources/stacks-chow/tags.txt, source lines 13902–13902. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04DG: sources/stacks-chow/tags.txt, source lines 5726–5726. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let V be a vector space. Let $Z \subset X \subset \mathbf{P}(V)$ be closed subvarieties with $Z \neq X \neq \mathbf{P}(V)$. Let $x \in Z$ be a point which is nonsingular on X. Then there is a nonempty Zariski open $U \subset \mathbf{P}(V)$ such that for all closed points $p \in U$ the morphism $r_p|_Z : Z \to r_p(Z)$ is an isomorphism over an open neighbourhood of $r_p(x)$.*

Proof. Arguing as in the proof of Lemma 0B1Q we find a nonempty Zariski open U such that for $p \in U$ the map $r_p|_X : X \to r_p(X)$ is étale at x and $r_p^{-1}(r_p(\{x\})) \cap Z = \{x\}$. Details omitted. Set $Z' = r_p(Z)$ viewed as a closed subvariety of $\mathbf{P}(W_p)$. Then the morphism $\pi : Z \to Z'$ is finite (Lemma 0B1P), unramified at x (small detail omitted) and $\pi^{-1}(\pi(\{x\})) = \{x\}$. The result follows from Étale Morphisms, Lemma 04DG. □

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